Outlook

Re: Data Request - Elephants evolved strategies reducing the biomechanical complexity of their trunk

From Michel C. Milinkovitch <Michel.Milinkovitch@unige.ch>
Date Tue 2025-09-16 12:00 PM
To Dagmar Fraser (Psychology) <d.s.fraser@bham.ac.uk>
Cc Jennifer Cook (Psychology) <j.l.cook@bham.ac.uk>

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Hi, you can now also find everything here:
https://zenodo.org/records/17094445
Best, M.

On 15 Sep 2025, at 11:43, Dagmar Fraser <d.s.fraser@bham.ac.uk> wrote:

Hey both,

Thanks ever so much for your quick response! I look forward to diving into your data.

Regards,

d

From: Michel C. Milinkovitch <Michel.Milinkovitch@unige.ch>
Sent: 15 September 2025 10:24 AM
To: Dagmar Fraser (Psychology) <d.s.fraser@bham.ac.uk>
Cc: Paule Dagenais <paule.dagenais@curie.fr>
Subject: Re: Data Request - Elephants evolved strategies reducing the biomechanical complexity of their trunk

CAUTION: This email originated from outside the organisation. Do not click links or open attachments unless you recognise the sender and know the c is safe.

Dear Dagmar,
Here is the link for you to download the data pertaining to 48 clean trajectories. 'Clean' means that Paule (in CC) extensively checked manually each marker (i.e., spurious signals have been removed).
https://drive.switch.ch/index.php/s/Wei9dlvjmyhabfv

You will see three folders:
-the trajectories using the ellipse parameters
-the trajectories using XYZ coordinates
-the videos (2 different angles of view for each trajectory)

The '**READ_ME_format**' file describes the structure of the trajectory files
We used 9 or 10 rows of markers on the trunk and 3 to 6 markers per row.
The frequency of registering the positions of the markers is 100 Hz.
The XYZ trajectories are straightforward but more cumbersome to use than the ellipse trajectories.

As described in our Current Biology paper (https://www.sciencedirect.com/science/article/pii/S0960982221011337 andhttps://www.lanevol.org/projects/proboscis), we fitted an ellipse to each row of markers and then used these ellipses to analyse the trajectories.
The '**READ_ME_format**' file indicates how the rotation matrix of each ellipse is computed in comparisons to an horizontal position of the ellipse with the large axis (a) pointing to the right (in the animal's reference coordinates), and the small axis (b) pointing forward.

We have more than 48 clean trajectories but ellipses were fitted on 'only' 48 of them.
The file '**READ_ME_sequences**' file lists all the clean trajectories we can provide you with (including the 48 that you have already access to).

Obviously, do not distribute or publish these data without our agreement.
We look forward to hearing from you.

Best regards,

Prof. Michel C. Milinkovitch
Laboratory of Artificial & Natural Evolution (LANE)
Dept of Genetics & Evolution, University of Geneva
& SIB (Swiss Institute of Bioinformatics)

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Our website: www.lanevol.org
News at: https://www.lanevol.org/news

On 14 Sep 2025, at 12:36, Dagmar Fraser <d.s.fraser@bham.ac.uk> wrote:

Dear Professor Milinkovitch,

I am a PhD student at the University of Birmingham with and interest in kinematic invariances of biological motion.

I write again in the hopes of your sharing of the elephant trunk kinematic data from "*Elephants evolved strategies reducing the biomechanical complexity of their trunk*".

We recently completed a review of power law analysis methods and would be delighted to explore your data with the new methods we proposed.

Looking forward to your response.

Regards,

d

Dagmar Scott Fraser (he/him)
Senior Centre Technician & Doctoral Researcher
Centre for Human Brain Health
University of Birmingham
School of Psychology
BlueSky / LinkedIn / X

From: Dagmar Fraser (Psychology) <d.s.fraser@bham.ac.uk>
Date: Friday, 6 June 2025 at 11:39
To: michel.milinkovitch@unige.ch <michel.milinkovitch@unige.ch>
Subject: Data Request - Elephants evolved strategies reducing the biomechanical complexity of their trunk

Hey,

We were utterly delighted with your article on elephant trunk kinematics. We are specifically interested in the velocity curvature power law, and it's calculations (see our most recent paper below).

We are preparing a followup paper with simulated data and whatever real world data we can get our hands on.

Would it be possible to access your qualisys data for the elephant trunk tip? We want to compare a variety of different analysis techniques to see if our vetted protocol for Power Law analysis stands up to scrutiny.

Thanks ever so much!

d

Dagmar Scott Fraser (he/him)
Senior Centre Technician & Doctoral Researcher
Centre for Human Brain Health
University of Birmingham
School of Psychology
BlueSky / LinkedIn / X

Recent Publications

Fraser, D. S., Di Luca, M., & Cook, J. L. (2025) Biological kinematics: a detailed review of the velocity-curvature power law calculation. *In Press, Experimental Brain Research* https://doi.org/10.31234/osf.io/vfq3d

Hickman, L. J., Sowden, S. L., **Fraser, D. S.**, Schuster, B. A., Rybicki, A., Galea, J. M., & Cook, J. L. (2024) Dopaminergic manipulations affect the modulation and meta-modulation of movement speed: evidence from two pharmacological interventions. *Behavioural Brain Research.* https://doi.org/10.1016/j.bbr.2024.115213

Schuster, B. A., Sowden, S. L., Rybicki, A. J., **Fraser, D. S.**, Press, C., Holland, P., & Cook, J. L. (2024) Disruption of dopamine D2/D3 system function impairs the human ability to understand the mental states of other people. *PLoS Biology*. https://doi.org/10.1371/journal.pbio.3002652

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